線性代數 期中考 (2025 Fall) — Eigen 部分

1. (10%) Find the eigenvalues and a basis for each eigenspace of $A = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$.

2. (15%) Let $A = \begin{pmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{pmatrix}$. (a) Find all eigenvalues. (b) Is A diagonalizable? Justify.

3. (10%) 是非題，並說明理由：

   a. If A is diagonalizable, then A is invertible.
   b. If A and B have the same eigenvalues, then A and B are similar.
   c. An eigenvector can be the zero vector.

4. (15%) Diagonalize $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$ if possible, and compute A^5.

5. (10%) Prove: if A is invertible and λ is an eigenvalue of A, then λ^{-1} is an eigenvalue of A^{-1}.

線性代數 期中考 (2024 Fall) — Eigen 部分

1. (10%) Find the characteristic polynomial and eigenvalues of $\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$.

2. (10%) True/False: A 3×3 matrix with 3 distinct eigenvalues is diagonalizable.

3. (15%) Let $A = PDP^{-1}$ where $P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, $D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. Compute A^4.

4. (10%) Prove that A and A^T have the same eigenvalues.